Arm® Cortex®-M33

32-bit Microcontroller

NuMicro® Family
M3351 Series BSP
Revision History

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Revision 3.00.003 (Released 2026-11-09)

- Support M3351xG.
- Add SBOM with fields of manufacturer and supplier.
- M3351.h adds RTC_NS definition.
- Sample code
  - ISP sample codes revise CMD_READ_CONFIG and CMD_UPDATE_CONFIG to support continuous read and user config update.
  - NuMaker-M3351SG: New.
- Third party
  - FatFs: Upgrade to R0.16.
  - FreeRTOS: Upgrade to v11.3.0.
- Document
  - VS Code Quick Start Guide adds a note: Must delete cbuild-pack yml file and NuMicro Family yaml file after updating pack.

Revision 3.00.002 (Released 2026-05-29)

- CMSIS upgrades to 6.3.0; CMSIS-Compiler upgrades to 2.2.0.
- Peripheral register
  - clk_reg.h: Add CLK_I3CHDRSCLEXT_HDRSCLEXTEN_Pos and CLK_I3CHDRSCLEXT_HDRSCLEXTEN_Msk.
  - i3c_reg.h: Add ADDRTBP and CHRTBP.
  - usbd_reg.h: Fix definitions of USBD_BUFSEG_BUFSEG_Msk and USBD_MXPLD_MXPLD_Msk.
- Standard driver
  - CANFD: Add CANFD_GetDataBitRate and CANFD_GetNominalBitRate.
  - CLK: Remove workarounds.
  - ECAP: Add ECAP_INPUT_MASK, ECAP_INTEN_MASK, ECAP_RLDEN_MASK and ECAP_STATUS_FLAG_MASK.
  - FMC: FMC_EraseBank replaces FMC_ISPCMD_BANK_ERASE.
  - I3C
    - I3C_CtrGetIBI improves Hot-Join stability.
    - I3C_DeviceInit enables HDR Mode SCL extended function.
    - I3C_TgtSend removes PDMA support.
  - SYS: SYS_BODCTL_BODVL_2_8V replaces SYS_BODCTL_BODVL_2_7V.
- Library
  - xmodem: Fix 128-byte unalignment send issue.
- Sample code
  - All sample code projects add no-unaligned-access compiler flag.
  - CANFD sample codes:
    - CANFD_BusOffRecovery: Remove.
    - CANFD_Loopback:Merged from CANFD_CAN_Loopback and CANFD_CANFD_Loopback with bus-off recovery handling.
    - CANFD_MonitorMode: Merged from CANFD_CAN_MonitorMode and CANFD_CANFD_MonitorMode with bus-off recovery handling.
    - CANFD_TxRx: Merged from CANFD_CAN_TxRx and CANFD_CANFD_TxRx with bus-off recovery handling.
    - CANFD_TxRxINT: Merged from CANFD_CAN_TxRxINT and CANFD_CANFD_TxRxINT with bus-off recovery handling.

- FMC_DualBankFwUpdate and FMC_FwUpdateApplication: Add CACHE_Flush after FMC_RemapBank.
- FMC_XOM, XOMLib, and XOMLibDemo: Refactor for dual bank.
- I3C_Controller
  - Add power-down and wake-up function.
  - Fix wrong access to Device Characteristics Table.
- I3C_Target: Add power-down and wake-up function.
- ISP_DFU: Remove dfu-util.
- TrustZone/HardFault: Remove workaround.

- Tool
  - HIDTransferTest
    - Move from USBD sample codes.
    - Add Control transfer.
  - mcuboot: Upgrade to v2.4.0 with imgtool.

Revision 3.00.001 (Released 2026-02-12)

- Startup
  - Improve stability of Reset_Handler with GCC compiler.
  - Remove unsafe printf.
- Peripheral register
  - sys_reg.h
    - SYS_SRAM_BISTSTS_SRBEND2_Pos and SYS_SRAM_BISTSTS_SRBEND2_Msk: Remove.
    - SYS_VREFCTL_PRELOAD_SEL_Pos and SYS_VREFCTL_PRELOAD_SEL_Msk: Fix definition.
- Standard driver
  - CLK
    - CLK_SetPowerDownMode: Add workaround.
    - CLK_SysTickDelay: SysTick uses current config if usable.
  - FMC: FMC_ISPSTS_INTFLAG_Pos definition changs to depend on secure state.
  - I3C: I3C_BusClkConfig improves stability of SDR read.
  - QSPI: QSPI_CLKSEL_PLL replaces QSPI_CLKSEL_PLL_DIV2.
  - SPI: SPI_CLKSEL_PLL replaces SPI_CLKSEL_PLL_DIV2.
  - SYS: Remove SYS_DISABLE_POR and SYS_ENABLE_POR.
- CMSIS driver
  - CAN, Flash, GPIO, I2C, SAI, SPI, USART, USBD, USBH, and VIO: Support.
- Sample code
  - All sample codes add IAR and VS Code projects.
  - CANFD_BusOffRecovery: Add.
  - I3C_Controller and I3C_Target: Update SDR SCL frequency.

Revision 3.00.000 (Released 2026-01-19)

- Initial release without IAR and VSCode projects.

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